
Research Proposal

Personalized and Privacy Aware Deep Learning using Split Learning and Moreau Envelopes

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Deep Learning has shown great promise in a number of domains, from healthcare, to public policy, to finance. In such applications, privacy is an important aspect to consider as data from various entities and sources is accumulated. With the growing array of devices that can collect data at high levels of granularity, personalization is another direction that has the potential to improve performance of deep learning systems. Statistical heterogeneity of data across clients and systems is one challenge that Federated Learning faces despite its privacy advantages, and personalization is thus an important aspect to consider.

This short proposal aims to address the following challenges:

- Traditional Federated Learning techniques do not take into account statistical heterogeneity of data and hence do not allow personalization.
- Techniques that do allow personalization come at significant computational costs at the client side.

Two perspectives are leveraged here to provide a potential solution to this problem: (i) Split Learning (Gupta and Raskar, 2018), and (ii) Similarity Regularization via Moreau Envelopes (T Dinh et al., 2020). Combining these would allow us to build deep learning systems that enable personalization, maintain privacy, and keep client computation in check. Figure 1 shows a preliminary visualization of the proposed approach.

Consider clients $A_1, A_2, \dots, A_N$ and a server S . The standard federated learning problem would be to minimize a loss function $f(w)$ over all clients, where w are weights of a common, global model:

$$\min_{w \in \mathbb{R}^d} \left\{ f(w) := \frac{1}{N} \sum_{i=1}^N f_i(w) \right\} \quad (1)$$

To introduce personalization as well as efficiency of computations, we can split computations between the clients and server and allow weights to be independent across clients while introducing a regularization term in the loss function to encourage similarity. Let weights of client i be denoted by θ_i , and weights of server by θ_{server} . The updated loss function is then:

$$\min_{w \in \mathbb{R}^d} \left\{ F(w) := \frac{1}{N} \sum_{i=1}^N F_i(w) \right\}, \text{ where } F_i(w) = \min_{\theta_i \in \mathbb{R}^d} \left\{ f_i(\theta_i, \theta_{server}) + \frac{\lambda}{2} \|\theta_i - w\|^2 \right\} \quad (2)$$

Hence, we encourage the local client weights θ_i to be *similar*, but not identical, to a parent model w , while the server side weights θ_{server} are common. Figure 1 shows a graphical depiction of the approach.

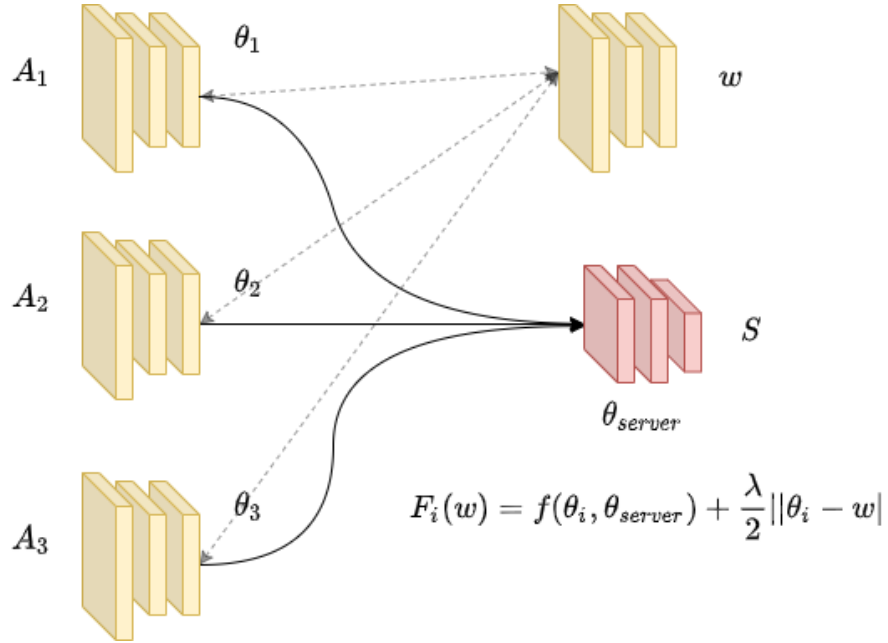


Figure 1. The proposed approach for personalized and privacy aware deep learning. A_1 , A_2 , and A_3 are clients, S is the server, θ_i are the weights for the corresponding client, θ_{server} are the weights for the server, w are weights of a reference global model, and $F_i(w)$ is the loss function of the i th client.

REFERENCES

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